MILITARY SPECIFICATION

CARTRIDGE, 5.56MM, BALL, XM777

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for fabricating and packing the Cartridge, 5.56MM, Ball, XM777.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 -	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 -	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-414 -	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-636 -	Visual Inspection Standards for Small Arms Ammunition through Caliber .50

FSC: 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-1168 - Lot Numbering of Ammunition

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND
PRODUCT AND PACKAGE DRAWINGS

11820420 - Cartridge, 5.56MM, Ball, XM777
7643674 - Classification of Cartridge Case Defects

PUBLICATIONS

SCATP-5.56 - Ammunition Ballistic Acceptance Test Methods, Test Procedures for 5.56MM Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM-E92 Method of test for Vickers Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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CODE OF FEDERAL REGULATIONS

TITLE 49 Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402. Orders should specify "49 CFR 100-199 (latest revision)"

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge. - The cartridge shall comply with all requirements specified on drawing 11820420, all associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Bullet extraction. - The force required to extract the bullet from the cartridge case shall not be less than 35 pounds.

3.4 Residual stress. - The cartridge case shall not split when subjected to a 1 percent mercurous nitrate solution for 15 minutes.

3.5 Waterproofness. - The cartridge shall not release more than one bubble of air when subjected to a positive internal pressure of 7.5 pounds per square inch (psi) minimum for 30 seconds minimum.

3.6 Accuracy. - The average of the mean radii of all targets of the sample cartridges shall not exceed 2.0 inches at 200 yards.

3.7 Velocity. - The average velocity of the sample cartridges, conditioned at $70^{\circ} \pm 2^{\circ}$ Fahrenheit (F), shall be 3175 feet per second (fps) plus or minus 40 fps, at 78 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 40 fps.

3.8 Chamber Pressure.

3.8.1 Measurement by piezoelectric transducer. - The average chamber pressure of the sample cartridges, conditioned at $70 \pm 2^{\circ}$F, shall not exceed 55,000 psi. The average chamber pressure plus three standard deviations of chamber pressure shall not exceed 61,000 psi.

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3.8.2 Measurement by copper-crush cylinder. - The average chamber pressure of the sample cartridges, conditioned at 70° ± 2°F, shall not exceed 52,000 psi. The average chamber pressure plus three standard deviations of chamber pressure shall not exceed 58,000 psi.

3.9 Port Pressure.

3.9.1 Measurement by piezoelectric transducer. - The average port pressure of the sample cartridges, conditioned at 70° ± 2°F, shall be 14,400 psi ± 2000 psi.

3.9.2 Measurement by copper-crush cylinder. - The average port pressure of the sample cartridges, conditioned at 70° ± 2°F, shall be 15,000 psi ± 2000 psi.

3.10 Temperature stability. - The pressure and velocity of sample cartridges conditioned and fired at the temperature extremes specified below shall be in accordance with 3.10.1, 3.10.2, and 3.10.3.

a) Conditioned at 125° ± 2°F for not less than one hour and fired at that temperature.

b) Conditioned at -65° ± 5°F for not less than one hour and fired at that temperature.

3.10.1 Velocity. - The average velocity, as determined in 3.7 above, shall not decrease by more than 250 ft/sec. Any increase in velocity is acceptable.

3.10.2 Chamber pressure. - The average chamber pressure, as determined by either method in 3.8 above, shall not increase by more than 5,000 psi. Any decrease in chamber pressure is acceptable.

3.10.3 Port pressure. - The average port pressure, as determined by either method in 3.9 above, shall neither increase nor decrease by more than 2,000 psi.

3.11 Function and casualty. - The cartridge shall function without casualty at ambient temperature and under the temperature conditions specified in 3.10.

3.12 Stripping. - Neither the jacket of the bullet, or any part thereof, nor the penetrator shall separate from the slug when the cartridge is fired.

3.13 Fouling. - The fouling accumulated in the weapon during the firing of 1000 sample cartridges shall not cause failure of the weapon to function.

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3.14 Penetration. - The cartridge shall demonstrate R100 penetration of 10 gauge (.135 inch) thickness SAE1010 or 1020 steel plate target with hardness between R_B55 minimum and R_B 70 maximum (NATO plate) positioned at 0 $\pm$ 5° obliquity and located 475 meters minimum from the muzzle of the weapon.

3.15 Workmanship. - The requirements for workmanship shall be as specified on applicable drawings, referenced specifications and in accordance with the following:

3.15.1 Metal Defects. - The cartridge shall be free of metal defects which includes but is not limited to: folds, wrinkles, scratches, scaly metal, dents, perforations, and other discontinuities.

3.15.2 Foreign matter. - The cartridge shall be free of corrosion, stains, discolorations, dirt, oil, and smears of lacquer.

3.15.3 Cleaning. - Cleaning methods used shall not be injurious to any part, nor shall the parts be contaminated by any cleaning agent.

3.15.4 Contamination of explosive components. - Extreme care shall be exercised to avoid contamination of primers and propellant by oil, grease, or other foreign matter.

3.16 Lot identification. - Each lot of ammunition shall be identified as to type, caliber, model, and lot number in accordance with MIL-STD-1168. In addition, each lot shall be identified by a Federal Stock Number assigned by the contracting officer.

3.17 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. - The following types of inspections shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Penetrator	11820426	50
Slug	11820427	25
Jacket	11820425	25
Bullet, Ball, 5.56MM (Steel Insert)	11820422	25
Cartridge Case (During Assy of primer)	10524200	25
Cartridge, 5.56MM, Ball, XM777	11820420	3500

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I of this specification.

4.3.3 Rejection. - See MIL-A-48078. Second samples and any other provisions for retest applicable to production lots shall not apply to the first article sample.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Cartridge and Components				See Below
					NEXT HIGHER ASSEMBLY
					See Below
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Penetrator (Dwg. 11820426) Examination for Defects	50		3.2	4.4.2.1
	Slug (Dwg. 11820427) Examination for Defects	25		3.2	4.4.2.2
	Jacket (Dwg. 11820425) Examination for Defects	25		3.2	4.4.2.3
	Bullet, Ball, 5.56mm (Steel Insert) (Dwg. 11820422) Examination for Defects	25		3.2	4.4.2.4
	Cartridge Case (During Assembly of Primer) (Dwg. 10524200) Examination for Defects	25		3.2	4.4.2.5

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Cartridge, 5.56MM, Ball, XM777				See Below
					NEXT HIGHER ASSEMBLY
					See Below
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge, 5.56MM, Ball, XM777 (Dwg 11820420)				
	Examination for defects	3500		3.2	4.4.2.6
	Bullet Extraction	25		3.3	4.5.2
	Waterproofness	50		3.5	4.5.4
	Residual Stress	50		3.4	4.5.3
	Ballistic Tests - See Table II				
	Accuracy	90		3.6	4.5.5
	Velocity (3 Temperatures)	60		3.7	4.5.6, 4.5.8 and Table II
	Chamber Pressure (3 Temperatures)	60		3.8	4.5.7, 4.5.8 and Table II
	Port Pressure (3 Temperatures)	60		3.9	4.5.7, 4.5.8 and Table II
	Function and Casualty	1440		3.11	4.5.9
	Stripping	100		3.12	4.5.10
	Fouling	1000		3.13	4.5.11
	Penetration	20		3.1.4	4.5.14

NOTES:

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of cartridges shall contain:

a. Primers from one lot interfix number from one manufacturer.

b. Propellant from not more than two lots and from one manufacturer.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling Plans (Attributes Data). - Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Sampling Plans (Variables Data). - When variables data is specified for a characteristic, sample size and accept-reject criteria shall be as specified for the double specification limit, variability unknown, standard deviation method in MIL-STD-414.

c. Acceptable Quality Levels (AQL's). - Examination for major and minor cartridge defects shall be performed on a class basis. The AQL for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non conforming cartridges shall be rejected and removed from the lot. Examination for major and minor component defects shall be performed on an individual AQL basis, using the AQL specified for each characteristic.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11820426
4.4.2.1	Penetrator				NEXT HIGHER ASSEMBLY 11820422
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Profile		0.40%	3.2	Gage
102.	Hardness (Variables data - 4.4.2b)		0.40%	3.2	4.5.13
103.	Weight (Variables data - 4.4.2b)		0.40%	3.2	Balance
Minor					
201.	Evidence of poor workmanship		1.0%	3.15	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

MIL-C-63352(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11820427
4.4.2.2	Slug				NEXT HIGHER ASSEMBLY 11820422
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major					
101.	Weight		0.40%	3.2	Balance
Minor:	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.3	Jacket		1 1	11820425
				NEXT HIGHER ASSEMBLY 11820422

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major					
101.	Wall thickness variation		0.40%	3.2	Gage
Minor:	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11820422
4.4.2.4	Bullet, Ball 5.56MM (Steel Insert)				NEXT HIGHER ASSEMBLY 11820420
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major					
101.	Length		0.40%	3.2	Gage
102.	Diameter		0.40%	3.2	Gage
103.	Weight (Variables data - 4.4.2b)		0.40%	3.2	Balance
104.	Boattail radius		0.40%	3.2	Gage
105.	Boattail length		0.40%	3.2	Gage
106.	Ogive contour		0.40%	3.2	Gage
107.	Cannelure width		0.40%	3.2	Gage
108.	Cannelure diameter		0.40%	3.2	Gage
109.	Distance from nose to cannelure		0.40%	3.2	Gage
110.	Diameter of point		0.40%	3.2	Gage
111.	Cannelure missing or damaged		0.40%	3.2	Visual
Minor					
201.	Workmanship		1.0%	3.15	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63352(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10524200
4.4.2.5	Cartridge Case (During assembly of Primer)				NEXT HIGHER ASSEMBLY 11820420
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Vent hole in cartridge case missing or obstructed.		100%	3.2	Automatic Gage
2.	Primer missing (32)		100%	3.2	Automatic Gage
3.	Primer cocked (33)		100%	3.2	Automatic Gage
4.	Primer inverted (34)		100%	3.2	Automatic Gage
Major					
101.	Vent hole diameter		0.40%	3.2	Gage
Minor:	None defined				

NOTES: NOTE: Numbers after defect descriptions refer to visual defect standards in MIL-STD-636 (NATO Caliber 7.62MM Section).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63352(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.6	Cartridge, 5.56MM, Ball, XM777				11820420
					NEXT HIGHER ASSEMBLY
					Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Perforated Case (6)		100%	3.2	Visual
2.	Case split in K, L, or M location (7)		100%	3.2	Visual
3.	Primer missing (32)		100%	3.2	Visual
4.	Primer cocked (33)		100%	3.2	Visual
5.	Primer inverted (34)		100%	3.2	Visual
6.	Weight min 1/ (See 4.4..2.b)		100%	3.2	Balance
Major	(See 4.4.2.c)				
101.	Total length			3.2	Gage
102.	Cartridge profile and alignment			3.2	Gage
103.	Diameter of extractor groove, max			3.2	Gage
104.	Diameter of head			3.2	Gage
105.	Thickness of head			3.2	Gage
106.	Length to shoulder datum			3.2	Gage
107.	Depth of primer			3.2	Gage
108.	Case split in I, S, or J location (6)			3.2	Visual
109.	Corrosion of stain with etching (2)			3.2	Visual
110.	Case head round (4)			3.2	Visual
111.	Beveled underside of head (10)			3.2	Visual
112.	Chamfer missing on head (rim) (13)			3.2	Visual
113.	Case dented (5)			3.2	Visual
114.	Case mouth not crimped in cannelure (11)			3.2	Visual
115.	No evidence of mouth anneal (21)			3.2	Visual

NOTES: Numbers after defect descriptions refer to visual defect standards in MIL-STD-636 (NATO Caliber 7.62MM Section)

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 11820420
4.4.2.6	Cartridge, 5.56MM, Ball, XM777				NEXT HIGHER ASSEMBLY Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116.	Scratch (8)			3.2	Visual
117.	Split bullet jacket (24)			3.2	Visual
118.	Loose bullet (25)			3.2	Visual
119.	Loose primer (35)			3.2	Visual
120.	Scaly metal (12)			3.2	Visual
Minor	(See 4.4.2.c)				
201.	Diameter of extractor groove, min			3.2	Gage
202.	Discolored, dirty, oily, or smeared (1)			3.2	Visual
203.	Case dented (5)			3.2	Visual
204.	Case scratched (8, 9)			3.2	Visual
205.	Scaly metal on case (12)			3.2	Visual
206.	Fold, wrinkle, buckle or buldge in case (14, 15, 16, 17)			3.2	Visual
207.	Head stamp missing or illegible (18)			3.2	Visual
208.	Defective head (19)			3.2	Visual
209.	Defective mouth (20)			3.2	Visual
210.	Bullet dented (22)			3.2	Visual
211.	Bullet scratched (23)			3.2	Visual
212.	Scaly metal on bullet (27)			3.2	Visual
213.	Upset (Crooked) point (28)			3.2	Visual
214.	Blunt point (30)			3.2	Visual
215.	Defective cannelure (31)			3.2	Visual
216.	Nicked or dented primer (36)			3.2	Visual

NOTES: Numbers after defect descriptions refer to visual defect standards in MIL-STD-636 (NATO Caliber 7.62MM Section)

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 11820420
4.4.2.6	Cartridge, 5.56MM, Ball, XM777				NEXT HIGHER ASSEMBLY Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
217.	No waterproofing material (primer pocket joint) (37)			3.2	Visual
218.	Defective crimp (38)			3.2	Visual
NOTE:	1/ Each lightweight cartridge shall be disassembled and the propellant weighed. Any cartridge containing less than 10 grains of propellant shall be classed as a critical defect. Any cartridge containing 10 grains or more of propellant shall be classed as a major defect.				

NOTES: Numbers after defect descriptions refer to visual defect standards in MIL-STD-636 (NATO Caliber 7.62MM Section)

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11820420
4.4.2.7	Cartridge, 5.56MM, Ball, XM777 (Testing)				NEXT HIGHER ASSEMBLY Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Bullet extraction		25-0-2		
102.	Waterproofness		75-1-2 50-0-3	3.3	4.5.2
103.	Residual stress in cartridge case (Mercurous Nitrate)		150-3-4 50-0-2	3.5	4.5.4
104.	Hardness of Cartridge Case Sidewall 1/	10	150-1-2	3.4	4.5.3
105.	Hardness of Cartridge Case Head at locations A, B, C, and D			3.2	4.5.12
106.	Ballistic Tests	See Table II and III	10-0-1	3.2	4.5.12
Minor:	None defined				
NOTE:	1/ Failure of the cartridge cases to comply with the applicable requirements shall be cause for rejection of the lot, subject to testing of a second sample consisting of double the quantity of cartridge cases in the first test. The cumulative results of all cartridge case in the combined first and second samples shall be used to determine compliance or noncompliance with the applicable requirements.				

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Packaging, Packing and Marking				See Section 5
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644. During or immediately prior to packaging, 100% inspection shall be performed to determine if the cartridge type conforms to the drawing. Occurrence of a high pressure test, dummy, or blank cartridge shall be classed as a critical defect. Occurrence of any type other than those listed shall be classed as a major defect. All nonconforming cartridges shall be rejected.				

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4.4.3 Testing. - Testing is described in the First Article Inspection (Table I) Quality Conformance Inspection tables (4.4.2.1 thru 4.4.2.8) and Table II. Firing defects and the association acceptance-rejection criteria are specified in Table III. Tests shall be conducted in accordance with the methods and procedures specified in 4.5.

4.4.3.1 Ballistic Test samples. - The quantities for the various ballistic tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the firing tests, and shall have been selected in a manner such that the sample is representative of the entire lot. Sufficient cartridges shall be selected such that all testing (including testing of second samples where applicable) can be performed. To assure a random sample for each test the sample cartridges from the lot shall be combined and intermixed before being divided into samples for the various tests.

TABLE II

Ballistic Test Sample Sizes

Test	Temperature				Req't Para	Test Para
	Ambient	70°+2°F	Low Temp (4.5.8.1)	High Temp (4.5.8.2)		
Accuracy 1/		150			3.6	4.5.5
Velocity 2/		50	50	50	3.7&3.10	4.5.6&8
Chamber & Port Pres 2/		50	50	50	3.8&3.9 &3.10	4.5.7&8 4.5.7
Function & Casualty 3/	480		480	480	3.11	4.5.9
Stripping 4/	100				3.12	4.5.10
Fouling 5/	1000				3.13	4.5.11
Penetration		20			3.14	4.5.14

1/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. The lot shall be rejected if any failing cartridges are found in the second sample.

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2/ Failure of the cartridges in any sample to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test for the temperature or temperatures at which the failure occurred. Both velocity and pressure shall be retested regardless of which one failed the first test. The lot shall be rejected if the cartridges in the second sample fail to comply with either the velocity or pressure requirements. Chamber pressure and port pressure tests shall be conducted simultaneously with velocity when the piezoelectric method is used to measure pressure.

3/ See Table III for defect classification and accept-reject criteria.

4/ The lot shall be rejected if two or more bullets from the sample cartridges are classed defective. If one defective bullet occurs in the first sample, a second sample consisting of double the quantity of cartridges in the first sample shall be tested. If any bullets in the second sample are classed defective the lot shall be rejected.

5/ The test shall be conducted on the first article sample only.

4.4.3.2 Accept-Reject Criteria for Function and Casualty Tests

The lot shall be rejected when function and casualty defects observed in all ballistic tests exceed the first sample reject number of Table III. If the number of defects found in the above tests exceeds the acceptance number for the first sample, but is less than the reject number for the first sample, a second sample, consisting of double the quantities specified under function and casualty test, shall be fired in the service weapon specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred in the first test. If the total number of defects in the combined first and second sample and firing tests exceeds the cumulative acceptance number, the lot shall be rejected. If, in testing a second sample defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

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4.4.3.3 Function and Casualty defects in ballistic tests. - For any ballistic test, except function and casualty, where the occurrence of a firing defect prevents the obtaining of a valid result for the characteristic being tested the following shall apply:

a) The defect shall be recorded under the appropriate function and casualty defect category and included in the defect count for determining acceptance or rejection in accordance with Table III.

b) The particular test for which the round was fired shall not be penalized.

c) A replacement round shall be fired to obtain the data for the characteristic being tested.

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TABLE III

FUNCTION AND CASUALTY DEFECT CLASSIFICATION

AND ACCEPT-REJECT CRITERIA FOR EACH TEMPERATURE

Defect Class	Defect Description	First Sample 480 Cartridges Accept No.	Reject No.	Cumulative Acceptance No. (1st & 2nd Sample) 1440 Total
Critical	Bullet remaining in bore 1	0	1	---
Critical	Misfire-No vent hole 6	0	1	---
Major	Misfire-Other cause 6	1	3	2
Major	Failure to extract	0	2	1
Major	Weapon Stoppage	0	2	1
	Primer Leak			
Major	a) Perforation of firing pin indent in primer cup	0	See 2/	None permitted in second sample
Major	b) Escape of gas through primer cup (excluding a. above	1	3	2
Minor	c) Escape of gas around primer cup 5	5	8	15
Major	d) Loose primer 7	1	2	2
Major	e) Blown primer 7	0	2	1
Major	f) Dropped primer 7	0	2	1
	Case Casualties 3			
	(a) Longitudinal Split			
Minor	(1) Neck and Shoulder (I or S)	5	8	15
Major	(2) Body (J)	2	5	4
Major	(3) Body(K)	0	2	1
Major	(4) To Head (L)	0	2	1
Major	(5) Through Head (M)	0	2	1
	(b) Circumferential Rupture			
Major	(1) Partial, shoulder or body (J & S)	1	3	2
Major	(2) Partial, body (K)	0	2	1
Major	(3) Partial, head (L)	0	2	1
Major	(4) Complete	0	2	1

1/ No second sample permitted. Lot shall be rejected if this defect occurs.

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2/ If one or more defects are found in the first sample, a second sample consisting of double the quantity of cartridges specified under Function and Casualty of Table II shall be fired. Prior to the testing of the second sample, the firing pin of the specific rifle(s) in which the defect originally occurred shall be replaced with a new firing pin. If an additional primer perforation is found in the second sample, the lot shall be rejected.

3/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

4/ All stoppages attributable to the ammunition, with the exception of misfire, complete rupture or failure to extract, observed in all tests shall be included.

5/ Of the cartridges in which this defect occurs, not more than 2 in the first sample nor more than 5 in the cumulative sample shall exhibit escape of gas around 50% or more than 50% of the periphery.

6/ Each cartridge that misfires shall be disassembled and examined for presence of vent hole . If the vent hole is missing, the lot shall be rejected with no second sample permitted.

7/ Primer Defect Descriptions

a. Loose primer. - Looseness, but not so as to permit the fired primer to fall from the primer pocket. The fired primer shall be classed as loose when (1) it protrudes above the primer pocket or (2) movement of the primer can be detected by pushing a probe, thumbnail, or fingertip against the primer.

b. Blown primer. - A primer which, when the cartridge is fired, is separated completely from the head of the cartridge case, and both the head of the case and the primer pocket are grossly distorted and deformed. The severity of this condition is such that it is readily detectable with the naked eye.

c. Dropped primer. - A primer which falls from the primer pocket after the cartridge is fired.

4.4.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 of this document.

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4.5 Test Methods and Procedures

4.5.1 Materials, Components, and Processes. - Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspections. - The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 and Table I herein.

4.5.2 Bullet extraction. - This test shall be conducted in accordance with SCATP-5.56. The rate of travel of the test head shall be not less than three nor more than six inches per minute. Any cartridge failing to comply with the applicable requirement shall be classed defective.

4.5.3 Residual stress in cartridge case. - This test shall be conducted in accordance with the mercurous nitrate test procedure specified in SCATP-5.56.

4.5.4 Waterproofness test of cartridge. - This test shall be conducted in accordance with SCATP-5.56.

4.5.5 Accuracy. - This test shall be conducted in accordance with SCATP-5.56. The mean radius shall be recorded to the nearest one-tenth of an inch.

4.5.6 Velocity. - This test shall be conducted in accordance with SCATP-5.56 and as specified in 4.5.8 below. Cartridges used for this test shall also be used for the chamber and port pressure measurements as specified in 4.5.7, when the piezoelectric method of pressure measurement is used.

4.5.7 Chamber and port pressures. - These tests shall be conducted simultaneously and in accordance with SCATP-5.56 and as specified in 4.5.8 below. Cartridges used for this test shall also be used for the velocity determination as specified in 4.5.6., when the piezoelectric method of pressure method is used.

4.5.8 High and Low Temperature. - These tests shall be conducted in accordance with SCATP-5.56. The weapon in which these tests are fired shall be at room temperature. Velocity and pressure differences shall be determined by conditioning fifty cartridges, from the same lot as the test cartridges, for 20 minutes minimum at $70^{o} \pm 2^{o}F$, and firing them immediately prior to firing the

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cartridges conditioned as specified below. All cartridges for the high and low temperature tests and for determining the velocity and pressure differences shall be from the same lot of ammunition.

4.5.8.1 Low Temperature Test. - The sample cartridges for this test shall be temperature conditioned at -65° ± 5°F for not less than 1 hour and shall be fired while at the conditioned temperature.

4.5.8.2 High temperature test. - The sample cartridges for this test shall be temperature conditioned at 125°F ± 2°F for not less than one hour and shall be fired while at the conditioned temperature.

4.5.9 Function and casualty. - This test shall be conducted in accordance with SCATP-5.56 and as specified in 4.5.8 above. Two rifles shall be used and the number of cartridges specified below shall be fired in each weapon:

Temperature	First Sample	Second Sample (if required)
Ambient	240	480
-65±5°F	240	480
+125±2°F	240	480

4.5.10 Bullet jacket stripping. - This test shall be conducted in accordance with SCATP-5.56 and the following: Five (5) magazines, each containing 20 cartridges, shall be fired alternately in the fully automatic and semi-automatic firing modes, i.e., first magazine is fired in the fully automatic mode, second magazine is fired in the semi-automatic mode, third magazine is fired in the fully automatic mode etc. After firing each magazine, the paper screen shall be inspected and replaced, as necessary, to assure each round perforates the paper. Each test weapon shall be cooled to ambient temperature after it has fired sixty (60) cartridges. The cooling cycle shall be the same for retests. Any bullet which shows evidence of stripping as indicated by irregular perforations on the paper screens, shall be classed as defective.

4.5.11 Fouling. - This test shall be conducted in accordance with SCATP-5.56.

4.5.12 Hardness test of cartridge case. - The cartridges to be subjected to this test shall have the bullets extracted, the propellant removed and the primers extracted. Each cartridge case shall then be prepared and tested in accordance with ASTM-E92.

4.5.12.1 Sidewall hardness. - The hardness of the case sidewall shall be determined at the locations specified on the applicable drawing. The readings shall be taken on the exterior surface and in approximately the same plane. The average hardness at each location shall be computed for the sample size specified on the drawing. The average hardness value at each location shall then be evaluated for compliance or non compliance with the maximum and minimum hardness requirements specified on the drawing.

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4.5.12.2 Head hardness. - After completion of the sidewall hardness determination the same cases shall be longitudinally sectioned and prepared for case head hardness testing. The hardness shall be determined at the four locations specified on the applicable drawing. If any hardness value at any point fails to comply with the minimum hardness requirement the corresponding point on the opposite side (remaining longitudinal section) of the primer pocket of the same head section from which the original value was obtained shall be tested. The higher of the two measurements shall be used for determining compliance with drawing requirements.

4.5.13 Hardness test of penetrator. - This test shall be conducted by sectioning the penetrator in half perpendicular to its longitudinal axis without altering its hardness characteristics. Hardness shall be determined at two loctions, 180° apart, on the sectioned surface for each penetrator. The two readings shall be located at a distance from the edge of the sectioned surface and from each other sufficiently so that the accuracy of the readings is maintained.

4.5.14 Penetration Test. - This test shall be conducted in accordance with SCATP-5.56. Any hit that demonstrates behind target damage as evidenced by examination of a 2024 T3 aluminum witness plate positioned parallel to and 6 inches minimum behind the target shall be considered to have penetrated the target plate.

5. PACKAGING

5.1 Packing - Level A (Worldwide shipment). - The cartridges shall be packed in accordance with the contract provisions, and in accordance with the Code of Federal Regulations, Title 49, Parts 100-199.

5.2 Marking and labeling. - Marking and labeling of the packed cartridges shall be in accordance with contract provisions and in accordance with the Code of Federal Regulations, Title 49 Parts 100-199.

5.3 Shipping. - When cartridges from more than one lot are shipped at one time the lots shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Intended Use. - The cartridges covered by this specification are intended for use in 5.56MM weapons.

6.2 Ordering Data. - See MIL-A-48078. In addition ordering data includes:

a. Type and level of packing for the cartridges.

b. Provisions for the submission of acceptance inspection results for each lot of ammunition presented to the Government.

6.3 Submission of Inspection Equipment Designs for Approval. - See MIL-A-48078. Submit inspection equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Hazard notice. - The cartridge described herein and some of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to prevent fire, explosion, adverse environment, rough handling, corrosive atmosphere, and electrically inducted incidents. Such measures shall include the employment of an effective safety program that addresses the inherent hazards associated with the cartridge.

6.5 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.6 Submission of test data. - In addition to the normal distribution of records, when the cartridge is procured by the US Army Armament Command, one (1) copy of all ballistic data and the ammunition data card for each lot shall be forwarded to: Commander, ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAA-SC.

Custodian:
ARMY-AR

Preparing Activity
ARMY-AR

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-63352 CARTRIDGE, 5.56MM, BALL, XM777

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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